

Yogendra Sangoju

AI/ML Engineer

yogiddtr@gmail.com | +1 551-229-2658 | Jersey City, NJ 07304

PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of experience designing and deploying scalable AI solutions in NLP and Computer Vision. Skilled in LLM fine-tuning (LoRA, QLoRA) and building production-ready ML systems using MLOps practices (CI/CD, Docker, FastAPI). Proven impact in optimizing AI workflows, including reducing model latency by 40% in healthcare applications through efficient cloud deployments on AWS and GCP.

EDUCATION

Master of Science (MS) - Computer Science Management,
Avila University

Kansas City, Missouri

Bachelor of Technology,
Jawaharlal Nehru Technological University (JNTUK)

Andhra Pradesh, India

SKILLS

Full-Stack Development: Full-stack development | Backend development | Frontend development | End-to-end feature development

Programming & Frameworks: TypeScript | NestJS | React

Databases & Caching: PostgreSQL | Redis

System Design & Architecture: System design | Clean architecture | Hexagonal architecture | Domain-Driven Design (DDD) | CQRS

API & Backend Systems: API design | Microservices | Backend architecture

Cloud & Infrastructure: Docker | GCP | Cloud Run | Pub/Sub | BigQuery | Terraform

AI & Advanced Systems: LLM integration | Embeddings | Vector databases

Data & Workflows: Data modeling | Data pipelines | Workflow orchestration (Temporal | Inngest)

Frontend Tools: TanStack | Tailwind CSS | Vite | shadcn/ui

Systems & Scalability: Scalable systems | Distributed systems | High-traffic systems

WORK EXPERIENCE

Infosoft, Junior Machine Learning Engineer

06/2023 – 12/2024 | New Jersey, USA

- Developed a predictive healthcare analytics platform using Python, SQL, and Machine Learning algorithms (Logistic Regression, Random Forest, XGBoost) to model diabetes patient readmission risk, applying statistical modeling, predictive modeling, and cross-validation to improve clinical prediction accuracy.
- Built Generative AI applications using Large Language Models (LLMs) by integrating Gemini APIs, implementing prompt engineering, embeddings, vector search, and Retrieval-Augmented Generation (RAG) pipelines to support patient queries on symptoms, medications, and treatment insights.
- Designed scalable LLM-based conversational systems and AI assistants, experimenting with agentic AI workflows, multi-agent orchestration, and structured outputs to enable contextual healthcare guidance.
- Implemented data preprocessing, feature engineering, dimensionality reduction (PCA), and SMOTE-based class imbalance handling using NumPy, Pandas, and SciPy, improving model evaluation metrics including precision, recall, F1-score, and ROC-AUC.
- Built interactive ML-driven dashboards and applications using Streamlit and React.js, visualizing predictions through Matplotlib, Power BI, and Tableau to support data-driven healthcare decisions.
- Deployed scalable ML services through REST APIs using FastAPI and Flask, containerizing applications with Docker and Kubernetes and deploying models on AWS and Google Cloud Platform (GCP) with CI/CD pipelines and experiment tracking.

Amazon, Software Development Engineer Internship

02/2023 – 05/2023 | Vijayawada

- Designed and developed scalable, cloud-native solutions using distributed systems and microservices architecture.
- Collaborated with cross-functional teams to build customer-focused features and contribute to technical design discussions.
- Wrote clean, maintainable code and participated in code reviews, ensuring best practices and high-quality standards.
- Worked in an Agile environment, contributing to CI/CD pipelines, testing, deployment, and operational support.
- Leveraged AI/ML and Generative AI tools to enhance development productivity and improve system performance.

- Developed end-to-end AI and machine learning systems using Python, Java, and SQL, designing data pipelines, ETL workflows, and backend system architectures for scalable ML applications.
- Built a BERT-based Natural Language Processing pipeline using PyTorch and Hugging Face Transformers to classify 10K+ monthly customer support tickets, applying deep learning, transformer architectures, and neural networks to improve resolution efficiency by 25%.
- Designed scalable REST API microservices using FastAPI and Flask, enabling real-time inference pipelines integrated with PostgreSQL, MySQL, and Redis databases.
- Implemented data preprocessing, feature engineering, model training, and GPU-accelerated training workflows, leveraging NumPy, Pandas, and Scikit-learn for model development and experimentation.
- Containerized ML applications using Docker, implemented CI/CD pipelines, and deployed production workloads on AWS and Microsoft Azure, ensuring scalable model deployment and monitoring.
- Worked with vector databases such as FAISS and Pinecone to enable semantic search, embeddings-based retrieval, and scalable AI-powered data processing systems.

PROJECTS & RESEARCH HIGHLIGHTS

A GENERATIVE AI ASSISTANT FOR DIABETES SUPPORT,

Domain: Healthcare AI / Generative AI / NLP

- Developed a Generative AI-powered healthcare assistant for diabetes management, enabling real-time symptom analysis, patient education, and personalized health insights through an interactive web platform.
- Built an LLM-based conversational chatbot (“Capsule”) using the Google Gemini API to handle natural language queries related to symptoms, medications, side effects, and treatment guidance.
- Implemented a scalable web application using Python, Streamlit, and React.js, and automated backend processes with Cron Jobs to support modular expansion such as Knowledge Center and recommendation modules.

AI-POWERED IMAGE-TO-VOICE NARRATOR,

Domain: Multimodal AI / Generative AI / Computer Vision

- Designed a multimodal Generative AI system that converts uploaded images into narrated short stories using computer vision, natural language generation, and speech synthesis.
- Implemented an image captioning pipeline using BLIP from Hugging Face, followed by story generation using OpenAI GPT-3.5 integrated via LangChain pipelines.
- Built and deployed an interactive Streamlit application on Streamlit Cloud and Hugging Face Spaces with secure environment configuration and modular pipeline architecture.

IMAGE CLASSIFICATION USING CNN IN PYTORCH, Domain: Computer Vision / Deep Learning

- Developed and trained a custom Convolutional Neural Network (CNN) using PyTorch for multi-class image classification tasks.
- Built efficient data preprocessing pipelines with Torchvision transforms and DataLoaders to handle image datasets and optimize training performance.
- Evaluated model performance using CrossEntropyLoss, Adam optimizer, and accuracy metrics, validating model generalization on unseen images.

PUBLICATIONS

- Internal White Paper: "TF-IDF-Based Ranking System for Doctor Recommendations"
- Internal Release: "Intent Detection and Classification in NLP Chatbots for Medical Triage"
- Google Cloud Certified - Professional Machine Learning Engineer
- AWS Certified Machine Learning - Specialty
- Natural Language Processing Specialization - DeepLearning.AI  (Coursera)
- Applied Data Science with Python - University of Michigan
- Deploying Machine Learning Models in Production
- SQL for Data Science - University of California, Davis
- Microsoft Exam AI-102: Designing and Implementing an Azure AI Solutions (preferred)